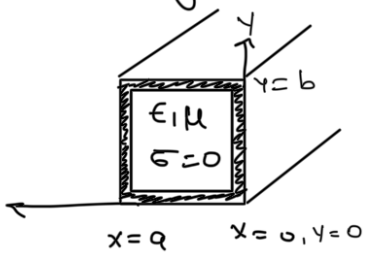
waveguide:

TL → GHz

microwave waveguide → 3GHz - 300GHz

optical waveguide → 200THz ⟶ There's a gap



$\sigma_c \Rightarrow \infty$

$\bar{\nabla}^2 \bar{H}_s + K^2 \bar{H}_s = 0$

$\bar{\nabla}^2 \bar{E}_s + K^2 \bar{E}_s = 0$

$\bar{\nabla} \times \bar{E} = -\frac{\partial \bar{B}}{\partial t} = -\mu \frac{\partial \bar{H}_s}{\partial t}$

$K = \omega\sqrt{\mu\epsilon}$ $\quad E = \bar{E}_s(x,y,z)e^{j\omega t}$

$\bar{\nabla} \times \bar{E}_s(x,y,z) = -j\omega\mu \bar{H}_s$

take curl

$$\frac{d^2E_{zs}(x,y,z)}{dx^2} + \frac{d^2E_{zs}(x,y,z)}{dy^2} + \frac{d^2E_{zs}(x,y,z)}{dz^2} + k^2E_{zs}(x,y,z) = 0$$

$$\frac{d^2E_{xs}(x,y,z)}{dx^2} + \frac{d^2E_{xs}(x,y,z)}{dy^2} + \frac{d^2E_{xs}(x,y,z)}{dz^2} + k^2E_{xs}(x,y,z) = 0$$

$$\frac{d^2E_{ys}(x,y,z)}{dx^2} + \frac{d^2E_{ys}(x,y,z)}{dy^2} + \frac{d^2E_{ys}(x,y,z)}{dz^2} + k^2E_{ys}(x,y,z) = 0$$

$E_{zs} \rightarrow X(x)\,Y(y)\,Z(z)$

$$\Rightarrow Y(y)Z(z)\frac{d^2X(x)}{dx^2} + X(x)Z(z)\frac{d^2Y(y)}{dy^2} + X(x)Y(y)\frac{d^2Z(z)}{dz^2} + k^2X(x)Y(y)Z(z) = 0$$

$$\Rightarrow \frac{1}{X(x)}\frac{d^2X(x)}{dx^2} + \frac{1}{Y(y)}\frac{d^2Y(y)}{dy^2} + \frac{1}{Z(z)}\frac{d^2Z(z)}{dz^2} = -k^2$$

$\frac{1}{X(x)}\frac{d^2X(x)}{dx^2} \rightarrow -k_x^2$ $\quad \frac{1}{Y(y)}\frac{d^2Y(y)}{dy^2} \rightarrow -k_y^2$ $\quad \frac{1}{Z(z)}\frac{d^2Z(z)}{dz^2} \rightarrow \gamma^2$

Standing wave

$$\Rightarrow \frac{d^2Z(z)}{dz^2} - \gamma^2 Z(z) = 0 \rightarrow Z(z) = C_5 e^{\gamma z} + C_6 e^{-\gamma z}$$

$$\Rightarrow \frac{d^2X(x)}{dx^2} + K_x^2 X(x) = 0 \rightarrow X(x) = C_1 \cos(K_x x) + C_2 \sin(K_x x)$$

$$\Rightarrow \frac{d^2Y(y)}{dy^2} + K_y^2 Y(y) = 0 \rightarrow Y(y) = C_3 \cos(K_y y) + C_4 \sin(K_y y)$$

$$R_x^2 + R_y^2 = h^2 = \gamma^2 + K^2 \qquad K^2 = \mu\omega^2\epsilon$$

$$E_{xs} = -\frac{\gamma}{h^2}\frac{dE_{zs}}{dx} - \frac{j\omega\mu}{h^2}\frac{dH_{zs}}{dy}$$

$$E_{ys} = -\frac{\gamma}{h^2}\frac{dE_{zs}}{dy} - \frac{j\omega\mu}{h^2}\frac{dH_{zs}}{dx}$$

$$H_{xs} = -\frac{j\omega\epsilon}{h^2}\frac{dE_{zs}}{dy} - \frac{\gamma}{h^2}\frac{dH_{zs}}{dx}$$

$$H_{ys} = -\frac{j\omega\epsilon}{h^2}\frac{dE_{zs}}{dx} - \frac{\gamma}{h^2}\frac{dH_{zs}}{dy}$$

$$E_{zs}(x,y,z) = (A_1 \cos K_x x + A_2 \sin K_x x)(A_3 \cos K_y y + A_4 \sin K_y y)e^{-\gamma z}$$

$$H_{zs}(x,y,z) = (B_1 \cos K_x x + B_2 \sin K_x x)(B_3 \cos K_y y + B_4 \sin K_y y)e^{-\gamma z}$$

(1) Transverse magnetic modes

$H_{zs} = 0 \quad E_{zs} \neq 0$

$E_{zs} = 0$ for $x = 0 ; x = a$, $y = 0 ; y = b$ } Boundary condition

$A_1 = A_3 = 0$

$$\Rightarrow E_{zs}(x,y,z) = E_0 \sin(K_x x)\sin(K_y y) \cdot e^{-\gamma z}$$

$(m, n = 0, 1, 2, \ldots)$

$\sin k_x a = 0 \Rightarrow k_x = \frac{m\pi}{a}$ $\quad m = 1, 2, \cdots$

$\sin k_y b = 0 \Rightarrow k_y = \frac{n\pi}{b}$ $\quad n = 1, 2, \cdots$

} (discrete) (unacceptable)

$m, n \in \mathbb{N}$

$$\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - \mu\omega^2\epsilon = \gamma^2$$

$\gamma \to \alpha + j\beta \Rightarrow$ Travelling wave

$$\gamma = \pm\sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - \mu\epsilon\omega^2}$$

$\gamma \to$ imaginary $\Rightarrow$ then only wave propogates (or else it just decays)

$$\omega_c^2 \mu\epsilon = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$ (we need $\omega > \omega_c$)

$\omega_c \to$ cut off frequency

wave guide $\to$ High pass filter

$$f_c = \frac{1}{2\sqrt{\mu\epsilon}}\left[\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2\right]^{1/2}$$

TM_{mn}

(TM_{11} shows lowest f_c)
